How does economic development influence environmental impact across countries and country groups ?

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Abstract

In this paper we investigate the link between economic development and environment across countries and country groups. Working on variable-income countries, we find that not all groups follow an inverted U-shape according to the model we use, so this hypothesis is not fully verified in contrast to a panel model that considers countries to be the same individuals over time. Electricity production from oil, gas and coal sources and the Foreign Direct Investment outflows (FDI) impact positively the emissions of CO2 emissions of countries and income countries where oil rents have a negative impact. Behind all these variations, which differ according to income group, there are explanations that we will try to explain and understand.

1 Introduction

"In 1950 the world emitted 6 billion tonnes of CO2. By 1990 this had almost quadrupled, reaching more than 20 billion tonnes. Emissions have continued to grow rapidly; we now emit over 35 billion tonnes each year. Emissions growth has slowed over the last few years, but they have yet to reach their peak" by Hannah Ritchie and Max Roser (2020) (18). By this rapid and huge growth of CO2 emissions over several years, which can be also proved with our data (see Figure 1); the environment and the ecological transition towards a more sustainable world have become one of the main topics discussed in the world in various ways, around meetings like the G20 or reports like the IPCC (Intergovernmental Panel on Climate Change).

Of all the possible explanations of an increase of CO2 emissions, one popular idea is that not all countries emit as much, and don't all have the same incentives and financial resources to reduce their emissions. Figure 2 and Figure 3 show heterogeneity between income groups: for several years, income group countries differ on their CO2 emissions. Understanding these differences is important and some literature explains well the inverted U-relationship between GDP per capita and CO2 emissions (i.e Figure 3 - high income countries). In fact, the EKC hypothesis stipulates that pollution increases because development progresses until they reach a "tipping point"; where as long as the country becomes developed it leads to reduction in emission (Grossman and Krueger, 1991 (10); Han et al., 2025(11)) . Our paper tests this hypothesis by adding two intermediate income groups, not only the low and high ones in order to extend the work already done. Figure 4 can give some ideas about this relationship: Upper and high income countries might follow a U-shaped relationship, while low income countries a N-shaped relationship. These observations will be proved by an econometric model in Section 3.

In terms of methods, our main econometric model is a two fixed effects model used to see the impact of each variable on CO2 emissions after controlling for country and year effects. It has been found that GDP per capita has a positive impact on CO2 emissions, but as it rises, emissions tend to fall, meaning that each group of countries follows an inverted U-shaped curve (Kuznets, 1955 (13); Grossman and Krueger, 1995 (10)). But we can see that the upper-middle countries pollute more than the high-income countries, because the latter invest more in renewable energies and in improving production technologies, whereas the poorer countries have more polluting industries and consume more fossil fuels. This may explain the negative effect of exports on CO2 emissions and the positive effect of electricity production from oil, gas and coal sources (Antweiler et al., 2001 (2)).

For all these reasons and because it must be common knowledge to understand the link between economic development and CO2 emissions, we will investigate: How does economic development influence environmental impact across countries and country groups ?

The remainder of this paper is organized as follows. Section 2 gives the presentation of the data and the institutional context. The econometric model and estimation are given in Section 3 . Section 4 provides the results of our models and then a conclusion of the project.

2 Data and Institutional Context

2.1 Data Presentation

We work with World Bank panel data containing 264 observations across countries and countries groups from 1960 to 2020. The World Bank traditionally collected data directly from member countries' systems (countries must first join the International Monetary Fund to become members). Quality of data collection is supported through collaboration with international organizations, including the United Nations (UN), the Organisation for Economic Co-operation and Development (OECD), the International Monetary Fund (IMF), and regional development banks, as well as through cooperation with donors.

We only focused on a few variables to concentrate our analysis and removed observations that were not countries (see Table A.1, Table A.2). Because the majority of the countries were already classified into one of the income groups of the World Bank, we also removed countries that were not in an income group. Moreover, we reduced our sample to only the 1980-2015 period in order to avoid a lot of missing values. To replace missing values, we used the k-NN (k-Nearest Neighbors) method, which is robust and suitable for non-linear variables and panel data. This method consists of replacing the missing values by the average of the observations closest in terms of Euclidean distance, in this case we used the five closest observations. In the end, we decided to exclude some countries that gave outlier values, specifically the top 3% of countries with the highest GDP per capita. At the end we left with 207 countries (see Table A.3).

2.2 Institutional Context

The 1980s began with a global recession linked to the second oil crisis, followed by an economic recovery. In the 1990s, globalisation and technological innovation accelerated sharply, leading to strong global growth until the major financial crisis in 2008. After this crisis, economic recovery was difficult, new economic powers emerged and inequalities increased. As far as the environment is concerned, warnings began to be sounded in the 1980s, followed by measures that remained limited. It was not until the 2010s, with the Paris Agreement in 2015, that drastic commitments were put in place. However, during this period, greenhouse gas emissions rose sharply, due in particular to the rapid growth of emerging economies.

2.3 Statistical Tables

In order to have a statistical approach on our model and understand some key things we describe countries characteristics by a descriptive statistics table (see Table 1). Because we focus on income groups countries, the same table can be found for each group. From Table 2 to Table 5, we can find respectively the countries of the Low Income group, the Lower middle income group, the Upper middle income group and the High income group. CO2 emissions vary enormously between countries, with some having very low levels in means, like low income countries (3.44 million tonnes of CO2 emissions) while others much more as high income countries (168.90 million tonnes of CO2 emissions) or the upper middle income countries (410.14 million tonnes of CO2 emissions). But the latter emits more than the high income which aligns with the work of Anderson et al. (2018)(1) : middle-income countries have much higher pollution levels than high income countries.

Moreover, the standard deviation for high income countries is 628.90 so 5 times greater than the average then it is shifted upwards because the range is so large between countries. This is key in our study, as we want to work on differences between countries' income groups, but one thing to bear in mind here is that they are observations for each country and a period of time together. Then, in order to understand these differences between income groups, we want to control for countries and year effect, see Section 3.

3 Model and Estimation

The objective of this paper is to estimate the effect of economic development on CO2 emissions across income groups. It can be affected by many variables but we will keep only the variables selected following the static analysis in Section 2. We have tested several regressions, retaining only the linear model by income group, which is interesting for comparing the differences between the control variables for different income groups, and the panel linear model, which is the most relevant. First, we performed the following tests to validate our two models.

3.1 Tests

3.1.1 VIf tests

The results presented in Table 6) may give cause for concern because of possible multicollinearity between the explanatory variables: the model explains almost 90% of the variation in log CO2. To verify this, we performed a VIF (Variance Inflation Factor) test on the fixed-effects model to check whether the variables are significantly related and validate the consistency of our estimators (Table 7) and Table 13)) respectively for the linear model by income group and the panel one. The conclusion of this test for both is that there is a strong multicollinearity between all variables. Nevertheless, the VIF test can be biased due to the presence of fixed effects and transformations on variables (i.e the log on variables, the squared or interaction term: that is exactly what we have). To check this issue again, we calculated the correlation matrix (including all the variables), see Table 8) and Table 14) , and performed a new VIF test (excluding the possible biased variables), see Table 9) and Table 15) . For the other model, we applied the same methodologies. These two methods showed us that all the variables have a negligible correlation for both models.

3.1.2 Hausman test

The Hausman Test is performed to test the hypothesis that it is preferable to use a random-effects model, assuming that country-specific differences are random and uncorrelated with the explanatory variables, while keeping the variables invariant over time for the panel model. The null hypothesis is : the random effect model is the efficient and consistent one because its estimators are not correlated with explanatory variables. In conclusion of this test we reject the null hypothesis (p-value < 0.001), so the fixed effects model is preferable. It corresponds to what we expected because of the heterogeneity of countries (see Table 18).

3.1.3 Wald tests

After validating the relevance of the fixed effects, we performed a robust Wald test on the panel linear model and a Wald test on the linear model (see Table 10 and Table 16), to assess the overall significance of the explanatory variable with the null hypothesis that all explanatory variables have no significant effect on the log(CO2 emissions) and the alternative hypothesis that there is at least one of explanatory variables that impact the log(CO2 emissions). The both tests result in the rejection of the null hypothesis with a p-value lower than 0.01. The fixed effects model has at least one of the explanatory variables that significantly impact the CO2 emissions.

3.2 Econometric models

3.2.1 Linear model by income groups

First, a linear regression for each group of countries is presented, including country and year fixed effects, respectively α_i and γ_t. Our regression includes a term grouping income categories. We estimate the following fixed effects model here to see the impact of each variable on log(CO2 emissions) once controlled by the country and year effects:

$$
\begin{aligned}
\log(\text{CO}_2 \text{ emissions}_{it}) = \alpha_i + \gamma_t + \beta_1 \log(GDP \text{ per capita}_{it}) + +\beta_2 \log(GDP \text{ per capita}_{it})^2 \\
+ \beta_3 Exports_{it} + \beta_4 ElectricityRenew_{it} \\
(1) \quad + [\beta_5 ElectricityFossil_{it} + \beta_6 FDIout_{it} + \beta_7 PopGrowth_{it} + \beta_8 OilRents_{it}] \\
+ \epsilon_{it}
\end{aligned}
$$

We chose the log for our dependent variable and for GDP per capita and its squared as explanatory variables because it reduces heterogeneity and therefore improves the efficiency of the estimates. To test whether the EKC hypothesis is validated or not, we include the GDP per capita and its square. Exports and electricity production from renewable sources are our other explanatory variables. And finally Oil rents, Electricity from oil, gas and coal, Population Growth and Foreign direct investment outflows are as controlled variables.

In fact, the country effect makes it possible to control for the specific and unobservable characteristics of each country, such as structural differences between countries, as Lutz Sager (2024) (20) proves, differences between countries can explain the main part of global pollution inequalities of countries. This control eliminates biases arising from characteristics such as geographical, climatic, political and institutional differences that vary over time (Brock and Taylor, 2003 (4); Torras and Boyce, 1998 (22); Khanna and Plassmann, 2006 (12)). Similarly, the fixed effect per year enables us to capture temporal trends common to all countries, such as economic crises.

3.2.2 Panel linear model

Additionally, captures heterogeneity and then improves the efficiency of estimator we decided to include a cross effect term between GDP and income group because as shown earlier in previous sections, CO2 emissions vary across income groups. This allow us to show you the second model of our paper, a panel linear model including this cross effect term :

$$\begin{aligned}
\log(\text{CO}_2 \text{ emissions}_{it}) = {} & \alpha_i + \gamma_t + \beta_1 \log(GDP \text{ per capita}_{it}) \\
& + \beta_2 \left(\log(GDP \text{ per capita}_{it}) \times IncomeGroup_{it}\right) \\
& + \beta_3 \ln(GDP \text{ per capita}_{it})^2 + \beta_4 Exports_{it} + \beta_5 ElectricityRenew_{it} \\
& + [\beta_6 ElectricityFossil_{it} + \beta_7 FDIout_{it} + \beta_8 PopGrowth_{it} + \beta_9 OilRents_{it}] \\
& + \epsilon_{it}
\end{aligned} \tag{2}$$

Here, the reference group in the Income Group category is “Low income”, simplifying our interpretations and comparisons of estimators.

4 Results

4.1 Linear model by income groups

In this model, we will concentrate our analysis on just some comparison between income groups with respect to some variables (see Table 11). First of all, only the Low-income and the High-income groups validate the EKC hypothesis significantly (with a GDP per capita positive and its square negatif) while the lower middle income group has a significant U-shaped relationship unlike the lower middle income its not significant.

Moreover, we observe that exports have a significant positive impact on CO2 emissions for low-income and lower-middle-income countries, compared to a negative effect for upper-middle-income countries. This means that exports in the upper-middle-income category reduce CO2 emissions by an average of 1.7% following a one-percentage-point increase in exports. This can be explained by the adoption of cleaner technology (Rafiq et al., 2016 (17)); Pablo-Romero et al., 2017 (16)), the development of the transport sector, or higher production costs, which may have led to the relocation of these industries to low-middle-income countries (Mills et Waite, (2009) (15); Rothman (1998) (19)) . Finally, high-income countries significantly reduce CO2 emissions by 0.9% following a one-percentage point increase in fossil fuels, demonstrating the energetic transition in these countries, confirming the work of Chakravorty et al. (2006 (5), 2008 (6)) and Boucekkine et al. (2013) (3).

4.2 Panel linear model

Our main model is the panel linear model (see Table 12). There are 6912 observations. The panel data model is balanced and explains around 87.5% of the variation of the logarithme of CO2 emissions. Because the adjusted R^2 is close to the R^2, the model is well specified and there is no overestimation of explanatory power due to too many variables.

We find that by controlling for country and year fixed effects, a one-percent increase in GDP per capita results in, an average, 2.09% significant rise in CO2 emissions ceteris paribus, and the slope of the GDP per capita is negative, indicating that as GDP increases, it has significatively less and less impact on the increase of CO2 emissions validating the EKC hypothesis, which gives different results from model 1. The three cross effect terms between each income group and the log of GDP per capita are positive and significant meaning that, in margin at constant country and year level , an increase of 1% in GDP per capita for high income countries leads to an increase of 0.806% in CO2 emissions more than for lower income countries (i.e 2.09%), which is less than for lower middle income countries (i.e.1.306%) and upper middle income countries (i.e 1.717%). The total effect for each income country is much higher, adding 2.095%, i.e on average, belonging to an upper-middle income group multiplies CO2 emissions ceteris paribus by almost 2 compared with the low income groups. The low margin effect on CO2 emissions of an increase in GDP per capita for high income countries compared to the other income countries can be explained by the fact that richer countries are adopting stricter environmental policies and cleaner technologies (Pablo-Romero et al., 2017 (16)).

There is also a link with the positive impact of the electricity production from oil, gas and coal sources in CO2 emissions. In fact, Stokey (1998) (21) and Charlier et al.(2024) (7) shows that in the first stages of development, countries polluting much more to create wealth but once they get sufficiently wealthy, they adopt cleaner technologies and pollution diminishes, validing once again the EKC hypothesis. Moreover, poor countries are more dependent on fossil energy. Then their impact of economic growth on pollution is higher than for rich countries where they invest on renewable energy production to reduce their emissions (Dong et al., 2018 (8)). In fact, our results show that a one-point increase in the share of total energy production from oil, gas and coal sources leads, controlling for country and year fixed effects, to an average 0.2% rise in CO2 emissions for all countries. It can be seen that the increase in CO2 emissions is significantly influenced by the consumption of fossil fuels. But by controlling the time and structural differences between countries, an increase of 1 point of percent on electricity production from renewable sources increases the emissions of CO2 by 0.3% on average. It has not a direct impact because it is not significant. So for that we can't say anything.

5 Conclusion

This study investigates how economic development influences environmental impact across countries and country groups. In fact, each country gives a huge importance to economic development, but in the last few years, the environment has also become another important factor to be taken into account when making certain production decisions, infrastructure or strategic means offered by international trade.

Nevertheless, the incentives to reduce CO2 emissions are not heterogeneous among countries, depending on their policy decisions and stage of development. That is what we wanted to study in our model. The EKC hypothesis is validated, meaning that depending on the income group, as countries get developed the emissions of CO2 decreases. Because richer countries are already well developed, they can invest in renewable sources and be less dependent on fossil fuel consumption in order to reduce their emissions and continue to have positive economic growth without harming the environment. Chakravorty et al. (2006 (5), 2008 (6)) and Boucekkine et al. (2013) (3). But some of these countries abuse their wealth and invest on pollutant infrastructure in low income countries in order to diminish their CO2 emissions while low income countries see theirs increase due to the lack of political power (Malm, 2016 (14)). These are some explanations of differences in CO2 emissions by income groups.

To have a better understanding of differences between economic development on CO2 emissions by income group some additional work can be done. In fact, taking a large sample of countries with more years can be done or including some period crises with no missing values in order to understand the impact of this chocs on the environmental trends. The use of k-NN method can bias our results in the sense that this method is very sensitive to neighbouring values, so the variance is very high due to the potentially large deviation of neighbouring observations, so it will be better to have no missing values or to have a better estimate of them. Moreover, changing from a country-year model to a more region-year-centered model can give better results and allows for a more detailed analysis of disparities within the same country. Finally, some important variables can be used in addition in order to make some policies and reglementation to limit the abusive actions by rich countries on poor countries (environmental policies) and to raise awareness of the importance of ecological transition for the well-being of all, both nationally and locally (life expectancy, premature mortality rate due to air pollution).

6 References

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A Appendix

Table A.1: Specifications of the Data Extract

Variable	Label	Selection
Country Name	Country Name	
Year	Year	1960–2020
Income Group	IncomeGroup	Low income, Lower middle income, Upper middle income, High income
Health Expenditures	Current health expenditure (% of GDP)	
Oil Rents	Oil rents (% of GDP)	
CO2 Emissions	CO2 emissions (kt)	
Renewable Energy Consumption	Renewable energy consumption (%)	
Electricity from Renewables	Electricity from renewables (% total, excl. hydro)	
Electricity from Fossil Fuels	Electricity from oil, gas, coal (% total)	
Urban Population Growth	Urban population growth (%)	
Population Growth	Population growth (%)	
FDI Inflows	Foreign direct investment, net inflows (% GDP)	
FDI Outflows	Foreign direct investment, net outflows (% GDP)	
GDP per Capita Growth (annual%)	GDP per capita growth (annual%)	
GDP per capita (constant 2010 US$)	GDP per capita (constant 2010 US$)	
GDP per capita (current US$)	GDP per capita (current US$)	
GDP growth (annual%)	GDP growth (annual%)	
GDP (constant 2010 US$)	GDP (constant 2010 US$)	
GDP (current US$)	GDP (current US$)	
Imports	Imports of goods and services (% GDP)	
Exports	Exports of goods and services (% GDP)	
Consumption Expenditure	Final consumption expenditure (% GDP)	
Population	Poptotals	

Notes: This table describe the specification of the data exctract from a World bank database. "Selection" refers to the cases selected at the extraction stage. This table only include the variables used for the work on this paper.

Table A.2: Sample Selection Procedures

Variable	Label	Restriction	Value Label
Year	Year	Keep	1980–2015
Income Group	IncomeGroup	Keep	Low income, Lower middle income, Upper middle income, High income
Country Name	Country Name	Keep	See Table A.3
Health Expenditures	Current health expenditure (% of GDP)	Keep	
Oil Rents	Oil rents (% of GDP)	Keep	
CO2 Emissions	CO2 emissions (kt)	Keep	
Renewable Energy Consumption	Renewable energy consumption (%)	Keep	
Electricity from Renewables	Electricity from renewables (% total, excl. hydro)	Keep	
Electricity from Fossil Fuels	Electricity from oil, gas, coal (% total)	Keep	
Urban Population Growth	Urban population growth (%)	Keep	
Population Growth	Population growth (%)	Keep	
FDI Inflows	Foreign direct investment, net inflows (% GDP)	Keep	
FDI Outflows	Foreign direct investment, net outflows (% GDP)	Keep	
GDP per Capita Growth (annual%)	GDP per capita growth (annual%)	Keep	
GDP per capita (constant 2010 US$)	GDP per capita (constant 2010 US$)	Drop	Top 3% highest values
GDP per capita (current US$)	GDP per capita (current US$)	Keep	
GDP growth (annual%)	GDP growth (annual%)	Keep	
GDP (constant 2010 US$)	GDP (constant 2010 US$)	Keep	
GDP (current US$)	GDP (current US$)	Keep	
Imports	Imports of goods and services (% GDP)	Keep	
Exports	Exports of goods and services (% GDP)	Keep	
Consumption Expenditure	Final consumption expenditure (% GDP)	Keep	
Population	Population (millions)	Keep	
CO2percap	CO2 emissions per capita	Computed	CO2 Emissions / Population
Missing Values	All variables	Imputed	k-NN (k=5)

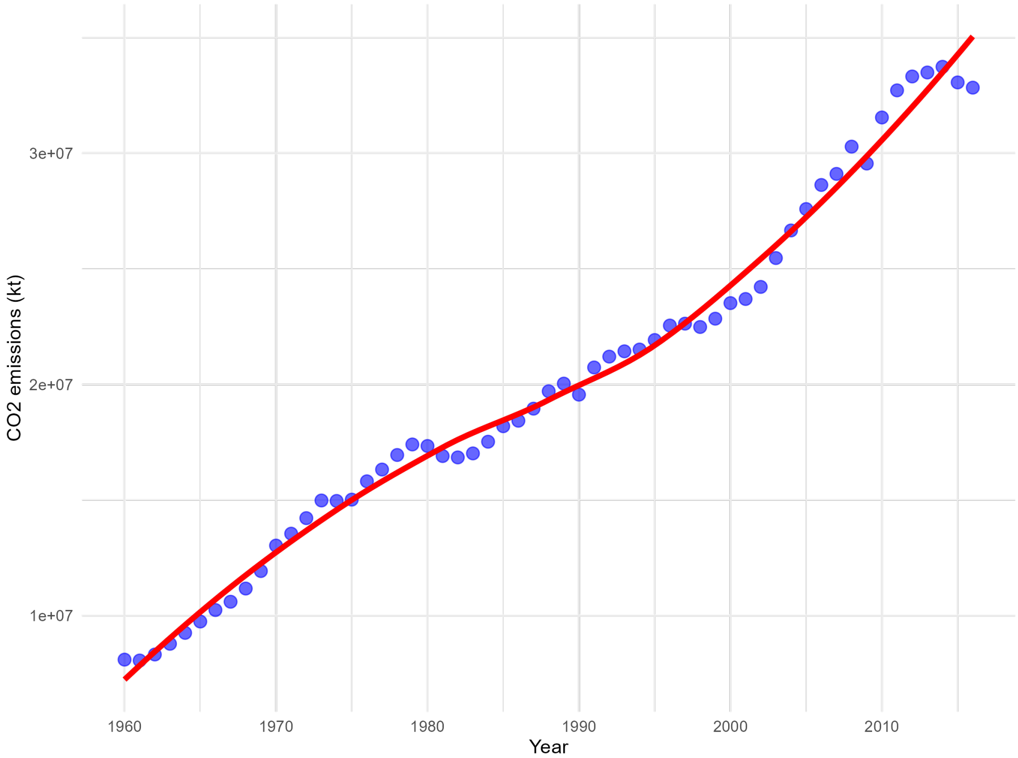
Notes: This table describes the sample selection procedures applied to refine the dataset. The original dataset covered the period 1960–2020, but was restricted to 1980–2015 to reduce missing values. Countries not classified into a World Bank income group were removed, as well as the top 3% of countries with the highest GDP per capita. Missing values were imputed using the k-Nearest Neighbors (k-NN) method. The variable "CO2 per capita" was created by dividing total CO2 emissions by the population of each country.

Table A.3: List of Countries Retained in the Sample

Country	Country	Country	Country
Afghanistan	Albania	Algeria	American Samoa
Andorra	Angola	Antigua and Barbuda	Argentina
Armenia	Aruba	Australia	Austria
Azerbaijan	Bahamas, The	Bahrain	Bangladesh
Barbados	Belarus	Belgium	Belize
Benin	Bhutan	Bolivia	Bosnia and Herzegovina
Botswana	Brazil	British Virgin Islands	Brunei Darussalam
Bulgaria	Burkina Faso	Burundi	Cabo Verde
Cambodia	Cameroon	Canada	Cayman Islands
Central African Republic	Chad	Channel Islands	Chile
China	Colombia	Comoros	Congo, Dem. Rep.
Congo, Rep.	Costa Rica	Croatia	Cuba
Cyprus	Czech Republic	Denmark	Djibouti
Dominica	Dominican Republic	Ecuador	Egypt, Arab Rep.
El Salvador	Equatorial Guinea	Eritrea	Estonia
Eswatini	Ethiopia	Faroe Islands	Fiji
Finland	France	French Polynesia	Gabon
Gambia, The	Georgia	Germany	Ghana
Gibraltar	Greece	Greenland	Grenada
Guam	Guatemala	Guinea	Guinea-Bissau
Guyana	Haiti	Honduras	Hong Kong SAR, China
Hungary	Iceland	India	Indonesia
Iran, Islamic Rep.	Iraq	Ireland	Isle of Man
Israel	Italy	Jamaica	Japan
Jordan	Kazakhstan	Kenya	Kiribati
Korea, Rep.	Kosovo	Kuwait	Kyrgyz Republic
Lao PDR	Latvia	Lebanon	Lesotho
Liberia	Libya	Liechtenstein	Lithuania
Luxembourg	Macao SAR, China	Madagascar	Malawi
Malaysia	Maldives	Mali	Malta
Marshall Islands	Mauritania	Mauritius	Mexico
Micronesia, Fed. Sts.	Moldova	Mongolia	Montenegro
Morocco	Mozambique	Myanmar	Namibia
Nauru	Nepal	Netherlands	New Caledonia
New Zealand	Nicaragua	Niger	Nigeria
North Macedonia	Northern Mariana Islands	Norway	Oman
Pakistan	Palau	Panama	Papua New Guinea
Paraguay	Peru	Philippines	Poland
Portugal	Qatar	Romania	Russian Federation
Rwanda	Samoa	San Marino	Saudi Arabia
Senegal	Serbia	Seychelles	Sierra Leone
Singapore	Sint Maarten (Dutch part)	Slovak Republic	Slovenia
Solomon Islands	Somalia	South Africa	South Sudan
Spain	Sri Lanka	St. Kitts and Nevis	St. Lucia
St. Vincent and the Grenadines	Sudan	Suriname	Sweden
Switzerland	Syrian Arab Republic	Tajikistan	Tanzania
Thailand	Timor-Leste	Togo	Tonga
Trinidad and Tobago	Tunisia	Turkey	Turkmenistan
Tuvalu	Uganda	Ukraine	United Arab Emirates
United Kingdom	United States	Uruguay	Uzbekistan
Vanuatu	Venezuela, RB	Vietnam	Virgin Islands (U.S.)
Yemen, Rep.	Zambia	Zimbabwe	

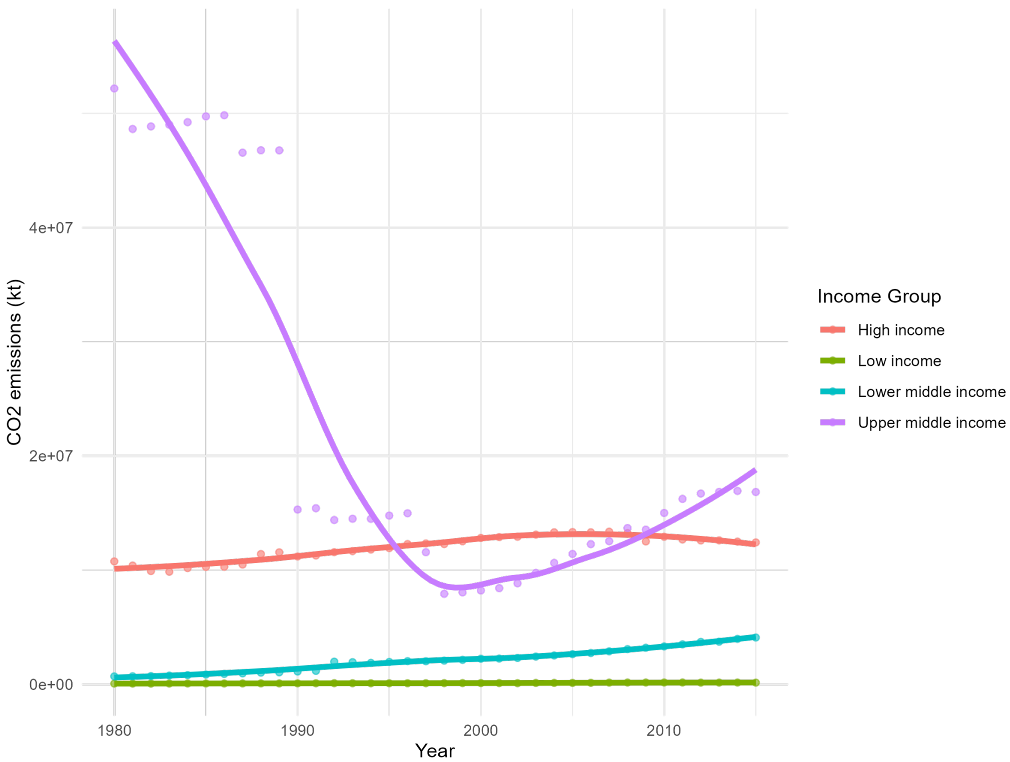
Notes: This table lists the 207 countries retained in the final sample after applying the selection criteria described in Table A.2.

Figure 1: World CO2 emissions (1960 - 2017)



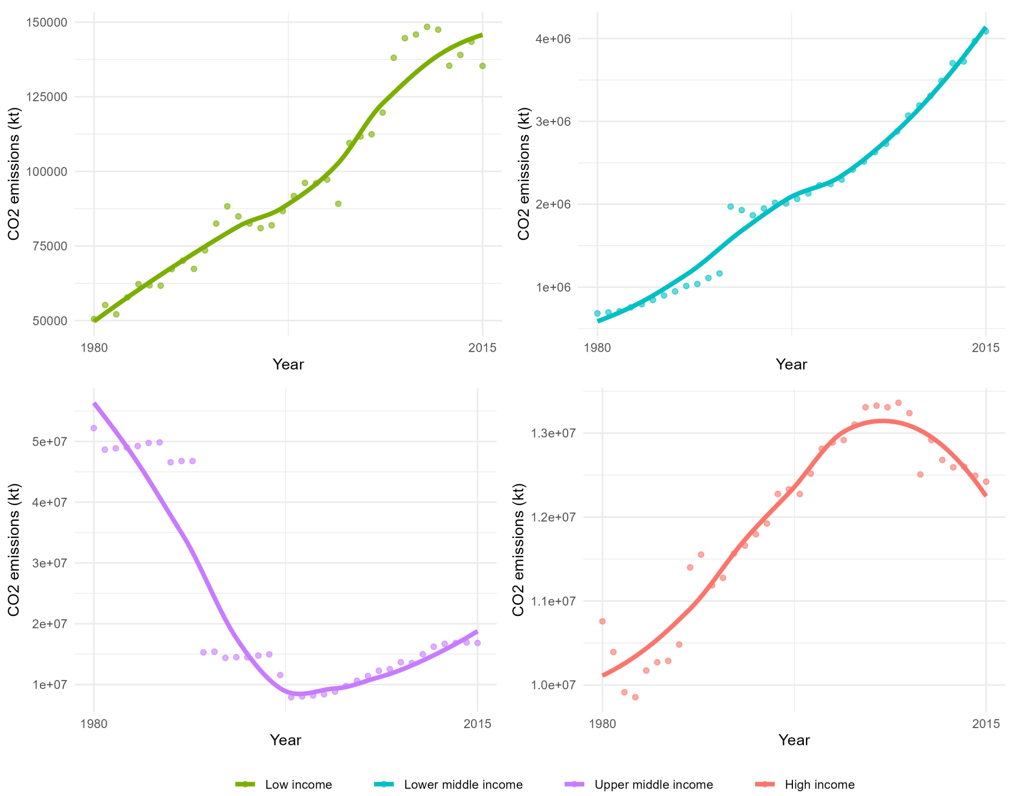
Note: Data source: World Bank. For this graph, we used our raw data before any modifications (i.e., without applying the KNN method, including outlier countries, including 1960-2017 time period).

Figure 2: CO2 emissions by Income Group



Note: Data source: World Bank. For this graph, we used our raw data before any modifications (i.e., without applying the KNN method, including outlier countries, including 1960-2017 time period).

Figure 3: CO2 emissions by Income Group (different scale for each group)



Note: Data source: World Bank. The graphs in this figure represent the same data as in Figure 2, but separated by income group for better readability. Each panel corresponds to a specific income group. The scales are differents depending on each income group.

Table 1: Descriptive statistics - 207 countries

Variable	mean	sd	1st Qu	median	3rd Qu	n
Health expenditures (% of GDP)	6.63	3.21	4.29	5.85	8.34	7264
Oil rents (% of GDP)	3.68	9.63	0.00	0.00	0.91	7264
CO2 emissions (kt, thousands)	183.89	725.35	0.66	5.27	51.51	7264
Renewable energy consumption (%)	31.86	32.39	2.52	19.13	56.60	7264
Electricity production from renewable sources (%)	1.84	5.19	0.00	0.00	0.87	7264
Electricity from oil, gas, and coal (%)	61.85	33.43	36.52	70.15	92.86	7264
Urban population growth (%)	2.50	2.28	0.89	2.20	3.85	7264
Population growth (%)	1.61	1.52	0.53	1.55	2.60	7264
GDP per capita growth (%)	2.05	5.84	0.12	2.55	4.24	7264
GDP per capita (constant 2010 US$, thousands)	9.56	12.89	1.38	4.17	10.65	7264
GDP per capita (current US$, thousands)	9.15	13.23	0.78	2.48	11.77	7264
GDP growth (%)	3.47	5.94	1.53	3.40	5.90	7264
Imports (% of GDP)	66.56	68.85	28.74	44.29	68.35	7264
Exports (% of GDP)	56.61	71.54	21.23	31.95	54.25	7264
Consumption expenditure (% of GDP)	81.70	18.47	71.56	80.27	91.06	7264
Foreign direct investment inflows (% of GDP)	51.34	138.77	0.50	2.02	6.16	7264
Foreign direct investment outflows (% of GDP)	4.01	16.71	0.00	0.10	0.89	7264
Population (millions)	28.83	115.44	0.75	5.20	16.75	7264
CO2 per capita (t)	0.76	16.23	0.00	0.00	0.01	7264

Notes: There are 207 countries in our sample. Data are from the World Bank for the period 1980-2015. These are panel data, we have 7264 observations in total. The sample exclude countries that does not belong to the one of the four income groups of the classification made by the world bank. Additionaly, outliers were removed, specifically the top 3% of countries with the highest GDP per capita, to ensure robustness in our findings.

Table 2: Descriptive statistics - Low income

Variable	mean	sd	1st Qu	median	3rd Qu	n
Health expenditures (% of GDP)	6.99	3.06	4.65	6.17	8.65	1008
Oil rents (% of GDP)	2.86	8.83	0.00	0.00	0.00	1008
CO2 emissions (kt, thousands)	3.44	8.61	0.54	0.96	2.26	1008
Renewable energy consumption (%)	76.5	27.2	76.4	87.5	93.2	1008
Electricity production from renewable sources (%)	0.34	0.69	0.00	0.00	0.68	1008
Electricity from oil, gas, and coal (%)	61.2	38.5	25.8	70.9	99.1	1008
Urban population growth (%)	4.39	2.09	3.4	4.26	5.37	1008
Population growth (%)	2.64	1.37	2.26	2.81	3.2	1008
GDP per capita growth (%)	1.08	6.22	-1.19	1.82	3.81	1008
GDP per capita (constant 2010 US$, thousands)	0.72	0.77	0.38	0.52	0.77	1008
GDP per capita (current US$, thousands)	0.59	1.26	0.26	0.39	0.73	1008
GDP growth (%)	3.54	6.32	1.37	4.39	6.50	1008
Imports (% of GDP)	42.69	33.59	26.13	32.96	44.91	1008
Exports (% of GDP)	25.06	25.11	12.69	17.80	28.40	1008
Consumption expenditure (% of GDP)	96.73	16.95	88.16	95.02	104.31	1008
Foreign direct investment inflows (% of GDP)	58.27	149.00	0.18	1.17	4.70	1008
Foreign direct investment outflows (% of GDP)	0.39	5.18	0.00	0.00	0.04	1008
Population (millions)	13.20	14.54	5.11	8.75	15.49	1008
CO2 per capita (t)	0.00	0.00	0.00	0.00	0.00	1008

Notes: There are 28 countries in this sample. We grouped the countries by income level from our initial sample of 207 countries and created a table for each income group. This table corresponds to the Low Income group.

Table 3: Descriptive statistics - Lower middle income

Variable	mean	sd	1st Qu	median	3rd Qu	n
Health expenditures (% of GDP)	5.38	2.65	3.70	4.50	6.27	1692
Oil rents (% of GDP)	3.23	8.22	0.00	0.00	1.43	1692
CO2 emissions (kt, thousands)	44.37	175.70	0.68	4.33	15.31	1692
Renewable energy consumption (%)	49.91	30.47	24.88	52.52	77.92	1692
Electricity production from renewable sources (%)	3.06	7.61	0.00	0.00	1.23	1692
Electricity from oil, gas, and coal (%)	51.62	34.19	16.61	55.25	84.87	1692
Urban population growth (%)	3.35	2.16	2.14	3.32	4.44	1692
Population growth (%)	2.05	0.99	1.42	2.24	2.74	1692
GDP per capita growth (%)	2.03	4.58	0.11	2.41	4.75	1692
GDP per capita (constant 2010 US$, thousands)	1.60	0.92	0.93	1.40	2.08	1692
GDP per capita (current US$, thousands)	1.13	0.90	0.55	0.81	1.37	1692
GDP growth (%)	4.00	4.60	2.13	4.41	6.44	1692
Imports (% of GDP)	43.99	24.23	27.29	37.97	58.15	1692
Exports (% of GDP)	28.92	16.62	18.41	25.04	37.77	1692
Consumption expenditure (% of GDP)	85.09	17.36	76.41	84.20	93.05	1692
Foreign direct investment inflows (% of GDP)	4.08	20.13	0.42	1.24	3.48	1692
Foreign direct investment outflows (% of GDP)	0.08	0.97	0.00	0.02	0.12	1692
Population (millions)	43.45	148.91	2.11	8.29	27.03	1692
CO2 per capita (t)	0.00	0.00	0.00	0.00	0.00	1692

Notes: There are 47 countries in this sample. We grouped the countries by income level from our initial sample of 207 countries and created a table for each income group. This table corresponds to the Lower Middle Income group.

Table 4: Descriptive statistics - Upper middle income

Variable	mean	sd	1st Qu	median	3rd Qu	n
Health expenditures (% of GDP)	6.42	3.20	4.30	5.60	7.93	2016
Oil rents (% of GDP)	4.68	10.28	0.00	0.04	3.54	2016
CO2 emissions (kt, thousands)	410.14	1134.31	1.56	15.86	129.92	2016
Renewable energy consumption (%)	20.66	20.33	2.88	14.80	33.22	2016
Electricity production from renewable sources (%)	1.01	2.92	0.00	0.00	0.30	2016
Electricity from oil, gas, and coal (%)	61.56	31.52	39.36	69.96	89.02	2016
Urban population growth (%)	2.21	2.00	0.93	2.04	3.20	2016
Population growth (%)	1.33	1.40	0.45	1.32	2.14	2016
GDP per capita growth (%)	2.48	8.12	-0.09	2.60	4.75	2016
GDP per capita (constant 2010 US$, thousands)	5.06	2.71	3.25	4.53	6.14	2016
GDP per capita (current US$, thousands)	3.33	2.89	1.23	2.42	4.51	2016
GDP growth (%)	3.69	8.29	1.22	3.30	6.04	2016
Imports (% of GDP)	50.31	28.08	27.12	49.05	64.36	2016
Exports (% of GDP)	37.29	21.03	23.97	30.64	45.91	2016
Consumption expenditure (% of GDP)	83.47	20.60	70.97	81.61	94.92	2016
Foreign direct investment inflows (% of GDP)	44.75	130.46	0.67	2.59	6.34	2016
Foreign direct investment outflows (% of GDP)	0.46	1.76	0.00	0.10	0.39	2016
Population (millions)	42.10	163.64	0.73	4.19	18.56	2016
CO2 per capita (t)	2.72	30.73	0.00	0.00	0.01	2016

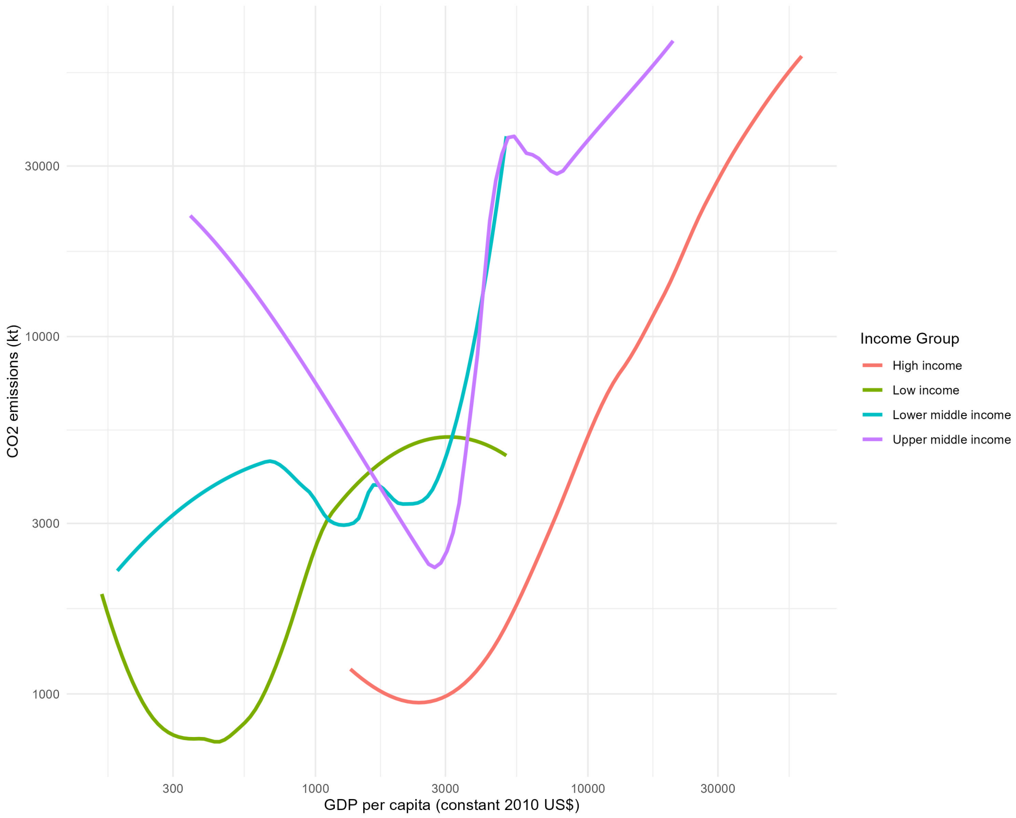
Notes: There are 56 countries in this sample. We grouped the countries by income level from our initial sample of 207 countries and created a table for each income group. This table corresponds to the Upper Middle Income group.

Table 5: Descriptive statistics - High income

Variable	mean	sd	1st Qu	median	3rd Qu	n
Health expenditures (% of GDP)	7.50	3.32	4.78	7.24	9.23	2548
Oil rents (% of GDP)	3.50	10.21	0.00	0.00	0.29	2548
CO2 emissions (kt, thousands)	168.90	628.73	0.67	16.39	78.00	2548
Renewable energy consumption (%)	11.05	14.74	0.08	4.76	17.00	2548
Electricity production from renewable sources (%)	2.27	5.37	0.00	0.02	1.65	2548
Electricity from oil, gas, and coal (%)	69.13	30.28	46.02	79.72	97.68	2548
Urban population growth (%)	1.41	1.91	0.34	1.01	1.93	2548
Population growth (%)	1.13	1.67	0.23	0.72	1.65	2548
GDP per capita growth (%)	2.09	3.93	0.65	2.60	3.75	2548
GDP per capita (constant 2010 US$, thousands)	21.90	15.03	7.37	18.83	33.56	2548
GDP per capita (current US$, thousands)	22.47	14.68	9.98	19.45	38.04	2548
GDP growth (%)	2.92	3.97	1.42	2.86	4.57	2548
Imports (% of GDP)	103.87	99.47	32.76	55.48	165.82	2548
Exports (% of GDP)	102.76	102.38	31.40	54.55	126.08	2548
Consumption expenditure (% of GDP)	72.09	11.49	63.91	74.40	79.70	2548
Foreign direct investment inflows (% of GDP)	85.19	172.39	0.73	2.70	12.07	2548
Foreign direct investment outflows (% of GDP)	10.86	26.65	0.18	1.43	7.13	2548
Population (millions)	14.81	37.91	0.15	2.55	10.30	2548
CO2 per capita (t)	0.01	0.02	0.00	0.01	0.01	2548

Notes: There are 76 countries in this sample. We grouped the countries by income level from our initial sample of 207 countries and created a table for each income group. This table corresponds to the High Income group.

Figure 4: CO2 emissions and GDP per capita by Income Group (log scale)



Note : Data source: World Bank. For this graph, we used our cleaned data after modifications (i.e., with applying the KNN method, excluding outlier countries, selecting 1980-2015 time period)

Table 6: Regression results - linear model

	Dependent variable: log(CO_2 emissions)
log(GDP per capita (constant 2010 US$))	1.017^{***}
	(0.044)
log(GDP per capita (constant 2010 US$))^2	-0.00004^{***}
	(0.00000)
Oil rents	-0.017^{***}
	(0.002)
Electricity production from renewable sources	-0.005
	(0.004)
Electricity from oil, gas and coal	-0.0004
	(0.001)
Foreign direct investment inflows	0.006^{***}
	(0.0001)
Population growth	0.061^{***}
	(0.012)
Exports	-0.006^{***}
	(0.0004)
Constant	-0.620^{*}
	(0.343)
Year fixed effects	Yes
Country fixed effects	Yes
Observations	6,912
R^2	0.899
Adjusted R^2	0.896
Residual Std. Error	0.909 (df = 6,677)
F Statistic	254.731^{***} (df = 234; 6,677)
Note:	*p<0.1; **p<0.05; ***p<0.01

Regression results of the linear model with country and year fixed effects.

Table 7: Variance Inflation Factors (VIF) - linear model

Variable	GVIF	Df	GVIF^(1/(2*Df))
GDP per capita (constant 2010 US$)	999.99	0.00	Inf
Oil rents	999.99	0.00	Inf
Electricity production from renewable sources	999.99	0.00	Inf
Electricity from oil, gas and coal	999.99	0.00	Inf
Foreign direct investment inflows	999.99	0.00	Inf
Population growth	999.99	0.00	Inf
Exports	4.57	1.00	2.14
Year	999.99	0.00	Inf
Country Name	999.99	0.00	Inf

Note: Extremely large values for certain variables have been replaced by 999.99 to improve table readability. VIF is biased due to the presence of transformed variables, fixed effects, and interaction terms.

Table 8: Correlation Matrix - linear model

	GDP per capita (constant 2010 US$)	Income Group numeric	Year numeric	Oil rents	Electricity from renewables	Electricity from oil, gas and coal	FDI outflows	Population growth	Exports
GDP per capita (constant 2010 US$)	1.00	-0.54	0.14	-0.03	0.14	0.05	0.10	-0.24	0.14
Income Group numeric	-0.54	1.00	0.00	0.07	-0.05	-0.10	-0.25	0.09	-0.35
Year numeric	0.14	0.00	1.00	0.02	0.15	-0.03	-0.02	-0.18	-0.06
Oil rents	-0.03	0.07	0.02	1.00	-0.12	0.12	-0.03	0.17	0.05
Electricity from renewables	0.14	-0.05	0.15	-0.12	1.00	-0.15	-0.03	-0.06	-0.09
Electricity from oil, gas and coal	0.05	-0.10	-0.03	0.12	-0.15	1.00	0.13	0.04	0.21
Foreign direct investment outflows	0.10	-0.25	-0.02	-0.03	-0.03	0.13	1.00	-0.11	0.55
Population growth	-0.24	0.09	-0.18	0.17	-0.06	0.04	-0.11	1.00	-0.05
Exports	0.14	-0.35	-0.06	0.05	-0.09	0.21	0.55	-0.05	1.00

Table 9: Variance Inflation Factors (VIF), corrected from bias - linear model

Variable	GVIF	Df	GVIF^(1/(2*Df))
Oil rents	1.68	0.00	Inf
Electricity production from renewable sources	1.68	0.00	Inf
Electricity from oil, gas and coal	1.68	0.00	Inf
Foreign direct investment inflows	1.68	0.00	Inf
Population growth	1.68	0.00	Inf
Exports	1.47	1.00	1.21

Note: VIF corrected from bias by excluding transformed variables, fixed effects, and interaction terms.

Table 10: Wald Test - Linear Model

Hypothesis
H0: All explanatory variable coefficients are equal to zero
H1: At least one coefficient is different from zero

Statistic	DF	P-Value
5724.898	234	< 2.2e-16

Note: The Wald test assesses whether the explanatory variables collectively have a significant effect on the dependent variable. Since the p-value is extremely small (< 0.01), we reject the null hypothesis (H_0), confirming that the model is jointly significant.

Table 11: Regression Results - linear model by Income Group

	Dependent variable: log(CO_2 emissions)			
	High Income	Upper Middle Income	Lower Middle Income	Low Income
	(1)	(2)	(3)	(4)
log(GDP per capita)	8.415^{***}	-5.903^{***}	0.080	3.376^{***}
	(0.731)	(1.122)	(0.727)	(0.349)
log(GDP per capita)^2	-0.450^{***}	0.447^{***}	0.033	-0.255^{***}
	(0.039)	(0.070)	(0.052)	(0.025)
Oil rents	0.021^{***}	-0.023^{***}	0.001	-0.011^{***}
	(0.003)	(0.004)	(0.004)	(0.002)
Electricity from renewables	-0.014^{***}	0.020	-0.007	0.036^{*}
	(0.004)	(0.015)	(0.004)	(0.020)
Electricity from oil, gas and coal	-0.009^{***}	0.010^{***}	0.0003	-0.001^{***}
	(0.001)	(0.002)	(0.001)	(0.0005)
Foreign direct investment inflows	0.001^{***}	0.010^{***}	-0.001	0.001^{***}
	(0.0002)	(0.0003)	(0.001)	(0.0001)
Population growth	-0.029^{**}	0.279^{***}	0.046	-0.040^{***}
	(0.011)	(0.030)	(0.034)	(0.009)
Exports	0.0005	-0.017^{***}	0.012^{***}	0.003^{***}
	(0.0004)	(0.002)	(0.002)	(0.001)
Constant	-32.235^{***}	27.114^{***}	7.121^{***}	-3.620^{***}
	(3.435)	(4.516)	(2.543)	(1.197)
Year fixed effects	Yes	Yes	Yes	Yes
Country fixed effects	Yes	Yes	Yes	Yes
Observations	2,196	2,016	1,692	1,008
R^2	0.967	0.897	0.914	0.917
Adjusted R^2	0.965	0.891	0.909	0.911
Residual Std. Error	0.534 (df = 2092)	1.055 (df = 1917)	0.708 (df = 1602)	0.376 (df = 937)
F Statistic	591.471^{***} (df = 103; 2092)	169.551^{***} (df = 98; 1917)	191.645^{***} (df = 89; 1602)	147.418^{***} (df = 70; 937)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table 12: Regression results - Panel linear model (fixed effects)

	Dependent variable: log(CO2 emissions)
log(GDP per capita (constant 2010 US$))	2.095***
	(0.442)
log(GDP per capita (constant 2010 US$))^2	−0.150***
	(0.032)
Oil rents	−0.021***
	(0.002)
Electricity production from renewable sources	0.003
	(0.004)
Electricity from oil, gas and coal	0.002***
	(0.001)
Foreign direct investment outflows	0.002**
	(0.001)
Population growth	0.080***
	(0.013)
Exports	−0.001
	(0.0004)
log(GDP per capita (constant 2010 US$)) × High income	0.806***
	(0.202)
log(GDP per capita (constant 2010 US$)) × Lower middle income	1.306***
	(0.118)
log(GDP per capita (constant 2010 US$)) × Upper middle income	1.717***
	(0.131)
Constant	0.174
	(1.542)
Year fixed effects	Yes
Country fixed effects	Yes
Observations	6,912
R^2	0.875
Adjusted R^2	0.871
Residual Std. Error	1.012 (df = 6674)
F Statistic	197.525*** (df = 237; 6674)
Note:	*p<0.1; **p<0.05; ***p<0.01

Regression results of the linear model with panel data, including country and year fixed effects.
Adding an interaction term between log(GDP per capita (constant 2010 US$)) and Income Group

Table 13: Variance Inflation Factors (VIF) - Panel linear model (fixed effects)

Variable	GVIF	Df	GVIF^(1/(2*Df))
GDP per capita (constant 2010 US$)	999.99	0.00	
Income Group	999.99	0.00	
Oil rents	-25.19	1.00	
Electricity production from renewable sources	-26.72	1.00	
Electricity from oil, gas and coal	-21.48	1.00	
Foreign direct investment outflows	-17.88	1.00	
Population growth	-0.68	1.00	
Exports	-21.03	1.00	
Year	999.99	0.00	
Country Name	999.99	0.00	

Note: Extremely large values for certain variables have been replaced by 999.99 to improve table readability. VIF is biased due to the presence of transformed variables, fixed effects, and interaction terms.

	GDP per capita (constant 2010 US$)	Income Group	Year	Oil rents	Electricity from renewables	Electricity from oil gas and coal	FDI Outflows	Population growth	Exports
GDP per capita (constant 2010 US$)	1.00	-0.54	0.14	-0.03	0.14	0.05	0.10	-0.24	0.14
Income Group	-0.54	1.00	0.00	0.07	-0.05	-0.10	-0.25	0.09	-0.35
Year	0.14	0.00	1.00	0.02	0.15	-0.03	-0.02	-0.18	-0.06
Oil rents	-0.03	0.07	0.02	1.00	-0.12	0.12	-0.03	0.17	0.05
Electricity from renewables	0.14	-0.05	0.15	-0.12	1.00	-0.15	-0.03	-0.06	-0.09
Electricity from oil, gas and coal	0.05	-0.10	-0.03	0.12	-0.15	1.00	0.13	0.04	0.21
Foreign direct investment outflows	0.10	-0.25	-0.02	-0.03	-0.03	0.13	1.00	-0.11	0.55
Population growth	-0.24	0.09	-0.18	0.17	-0.06	0.04	-0.11	1.00	-0.05
Exports	0.14	-0.35	-0.06	0.05	-0.09	0.21	0.55	-0.05	1.00

Table 14: Correlation Matrix - panel data

Table 15: Variance Inflation Factors (VIF), corrected from bias - Panel linear model (fixed effects)

Variable	GVIF
Oil rents	1.06
Electricity production from renewable sources	1.04
Electricity from oil, gas and coal	1.08
Foreign direct investment outflows	1.46
Population growth	1.04
Exports	1.49

Note: VIF corrected from bias by excluding transformed variables, fixed effects, and interaction terms.

Table 16: Robust Wald test - Panel linear model (fixed effects)

Hypothesis
H0: The model has no joint significance
H1: The model has joint significance

Statistic	DF	P-Value
166.976	11 & 191	1.588e-07

Note: The robust Wald test assesses whether the independent variables collectively have a significant effect on the dependent variable. Also the test is robust to heteroskedastic errors. Since the p-value is very small (< 0.01), we reject the null hypothesis (H_0), confirming that the model is jointly significant.

Table 17: Comparison of Fixed Effects (FE) and Random Effects (RE) Panel Models

	Dependent variable: log(CO2 emissions)	
	Fixed Effects (FE)	Random Effects (RE)
	(1)	(2)
Income Group (reference: Low Income)		
High income		-4.801^{***}
		(1.694)
Lower middle income		-8.651^{***}
		(0.936)
Upper middle income		-11.810^{***}
		(1.072)
log(GDP per capita (constant 2010 US$))	2.095^{***}	1.820^{***}
	(0.442)	(0.441)
(GDP per capita (constant 2010 US$))^2	-0.150^{***}	-0.126^{***}
	(0.032)	(0.032)
Oil rents	-0.021^{***}	-0.020^{***}
	(0.002)	(0.002)
Electricity from renewables	0.003	0.006
	(0.004)	(0.004)
Electricity from oil, gas and coal	0.002^{***}	0.002^{***}
	(0.001)	(0.001)
Foreign direct investment outflows	0.002^{**}	0.003^{***}
	(0.001)	(0.001)
Population growth	0.080^{***}	0.061^{***}
	(0.013)	(0.013)
Exports	-0.001	-0.001
	(0.0004)	(0.0004)
Interaction Terms		
log(GDP per capita) × High income	0.806^{***}	0.817^{***}
	(0.202)	(0.202)
log(GDP per capita) × Lower middle income	1.306^{***}	1.356^{***}
	(0.118)	(0.116)
log(GDP per capita) × Upper middle income	1.717^{***}	1.717^{***}
	(0.131)	(0.131)
Constant		0.420
		(1.580)
Observations	6,912	6,912
R^2	0.099	0.124
Adjusted R^2	0.067	0.123
F Statistic	66.429^{***} (df = 11; 6674)	980.181^{***}

Note: *p<0.1; **p<0.05; ***p<0.01.
Comparison of FE and RE models. The FE model controls for time-invariant characteristics. The factor variable "Income group" is only present in the RE model because the FE model controls for heterogeneity between income groups. These factor variables are absorbed by the fixed country effect in the FE model.

Table 18: Hausman Test: Fixed Effects vs. Random Effects

Hypothesis
H0: Random Effects are appropriate
H1: Random Effects are not appropriate

Statistic	DF	P-Value
166.976	11	<2e-16

Note: The Hausman test compares the Fixed Effects (FE) and Random Effects (RE) models. Since the p-value is very small (< 0.01), we reject the null hypothesis (H_0), indicating that the Fixed Effects model is preferred. This suggests that unobserved individual effects are correlated with the regressors, the FE model is the preferable choice